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AIM

Building a Simple Data Science Pipeline: Integrate all learned techniques in previous labs into a complete workflow.

THEORY

1. What is a Data Science Pipeline? Explain its key stages with examples.

A Data Science pipeline is a sequence of steps used to collect, process, analyze, and build models from data to solve a problem.

Key Stages:

Data Collection: Gathering data (e.g., from databases, APIs, sensors). Example: Collecting sales data from an e-commerce website.

Data Cleaning: Handling missing values, removing duplicates. Example: Filling missing prices with average values.

Exploratory Data Analysis (EDA): Understanding patterns and relationships. Example: Visualizing sales trends.

Feature Engineering: Creating useful input features. Example: Extracting "month" from date.

Model Building: Applying ML algorithms. Example: Using regression to predict sales.

Evaluation: Checking model accuracy. Example: Using metrics like accuracy or RMSE.

Deployment: Using the model in real-world applications.

2.What is Exploratory Data Analysis (EDA)?

EDA is the process of analyzing datasets to summarize their main characteristics using statistics and visualizations.

Univariate Analysis:

Analyzes one variable at a time Helps understand distribution, mean, median
Example: Histogram of age

Bivariate Analysis:

Analyzes relationship between two variables Helps find correlation or patterns
Example: Scatter plot of height vs weight

3. Differentiate between Label Encoding and One-Hot Encoding. When should each be used?

Feature	Label Encoding	One-Hot Encoding
Method	Assigns numbers to categories	Creates binary columns
Example	Red=0, Blue=1, Green=2	Red=[1,0,0], Blue=[0,1,0]
Use Case	Ordinal data (ordered)	Nominal data (no order)
Risk	May create false order	No order assumption

When to use:

Use Label Encoding → when categories have order (e.g., Low, Medium, High)

Use One-Hot Encoding → when categories are independent (e.g., colors)

4. Why is feature scaling required? Compare normalization and standardization.

Feature Scaling is a technique used to bring all input features to a similar scale so that no single feature dominates the model due to its larger numerical value.

Why Feature Scaling is Required:

Many machine learning algorithms (like KNN, SVM, and Gradient Descent-based models) are sensitive to feature magnitudes. Features with larger values can dominate smaller ones and bias the model. It improves model accuracy and speeds up convergence during training.

Comparing Normalization and standardization

Feature	Normalization	Standardization
Range	[0, 1]	No fixed range
Based on	Min and Max values	Mean and Standard deviation
Outlier Sensitivity	High	Low
Usage	When data is bounded	General-purpose scaling

5.What is feature engineering? Explain how creating new features can improve model performance

Feature Engineering is the process of creating new input features or transforming existing features in a dataset to improve the performance of machine learning models.

It involves using domain knowledge and data analysis techniques to make the data more meaningful and suitable for modeling.

Examples:

Extracting day/month/year from date
Creating BMI = weight/height²
Combining features (e.g., total_price = quantity × price)

How Feature Engineering Improves Model Performance:

Reveals hidden patterns: New features can highlight relationships that were not obvious in raw data

Improves model accuracy: Better input features help the model learn more effectively

Reduces complexity: Meaningful features make it easier for the model to understand the data

Handles non-linearity: Transformations (like square or log) help models capture complex relationships

```
In [ ]: #Step 1: Import Libraries
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_r

# Step 2: Load Dataset (Seaborn built-in dataset)
df = sns.load_dataset('titanic')

# Step 3: Basic Inspection
print(df.head())
```

Enhances generalization: Well-designed features help the model perform better on unseen data

```

print(df.info())
print(df.isnull().sum())

# Step 4: Data Preprocessing

# Select useful columns

df = df[['survived', 'pclass', 'sex', 'age', 'fare', 'embarked']]

# Handle missing values
imputer = SimpleImputer(strategy='mean')
df['age'] = imputer.fit_transform(df[['age']])

df['embarked'].fillna(df['embarked'].mode()[0], inplace=True)

# Encode categorical variables
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])
df['embarked'] = le.fit_transform(df['embarked'])

# Step 5: Exploratory Data Analysis (EDA)

# Survival count
sns.countplot(x='survived', data=df)
plt.title("Survival Count")
plt.show()

# Age distribution
sns.histplot(df['age'], kde=True)
plt.title("Age Distribution")
plt.show()

# Step 6: Feature Scaling
scaler = StandardScaler()
df[['age', 'fare']] = scaler.fit_transform(df[['age', 'fare']])

# Step 7: Split Data
X = df.drop('survived', axis=1)
y = df['survived']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Step 8: Model Building
model = LogisticRegression()
model.fit(X_train, y_train)

# Step 9: Prediction
y_pred = model.predict(X_test)

# Step 10: Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))

```

```
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

	pclass	survived	sex	age	sibsp	parch	fare	embarked	class	\
0	3	0	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	3	1	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	3	0	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 15 columns):

#	Column	-----	Non-Null Count	---	Dtype	----
---	survived		891 non-null	-	int64	
0	pclass		891 non-null		int64	
1	sex		891 non-null		object	
2	age		891 non-null		float64	
3	sibsp		891 non-null		int64	
4	parch		891 non-null		int64	
5	fare		891 non-null		float64	
6	embarked		891 non-null		object	
7	class		891 non-null		category	
8	who		891 non-null		object	
9	adult_male		891 non-null		bool	
10	deck		891 non-null		category	
11	embark_town				object	
12	alive				object	
13	alone				bool	
14						

dtypes: bool(2), category(2), float64(2), int64(4), object(5) memory

usage: 80.7+ KB

None

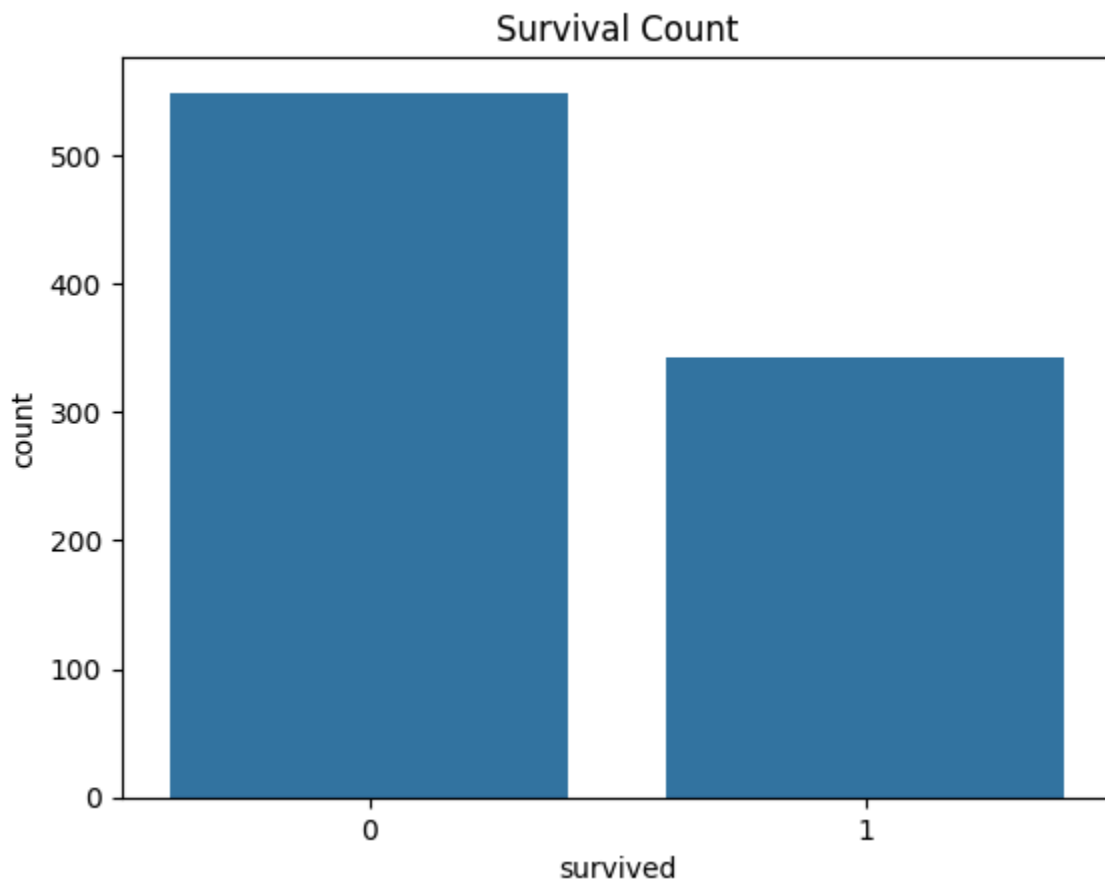
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

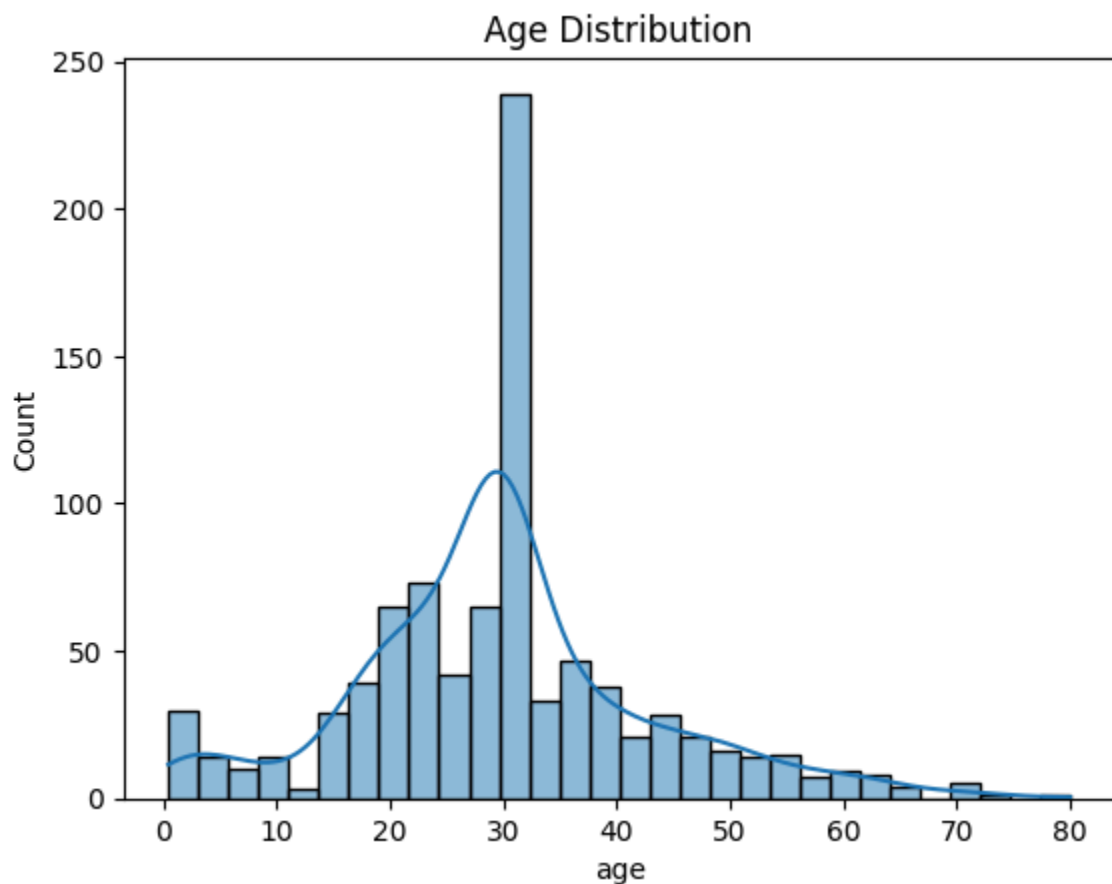
dtype: int64

/tmp/ipykernel_2619/3761747590.py:30: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['embarked'].fillna(df['embarked'].mode()[0], inplace=True)
```





Accuracy: 0.7988826815642458

Confusion Matrix:

```
[[88 17]
 [19 55]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.82	0.84	0.83	105
1	0.76	0.74	0.75	74
accuracy			0.80	179
macro avg	0.79	0.79	0.79	179
weighted avg	0.80	0.80	0.80	179

Conclusion

In this experiment, we studied the complete data science pipeline, including data preprocessing, exploratory data analysis (EDA), feature encoding, feature scaling, and feature engineering. We understood how proper data preparation and transformation techniques improve the quality of data and enhance model performance.

EDA helped in identifying patterns and relationships, while encoding and scaling ensured that the data was suitable for machine learning algorithms. Feature engineering further improved the model by creating meaningful features from raw data.

Overall, the experiment demonstrated that well-prepared data and effective feature design play a crucial role in building accurate and efficient machine learning models.